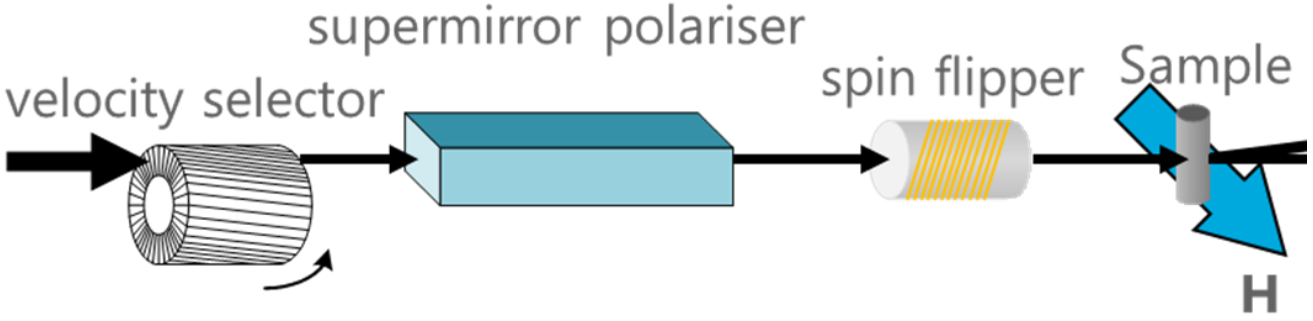
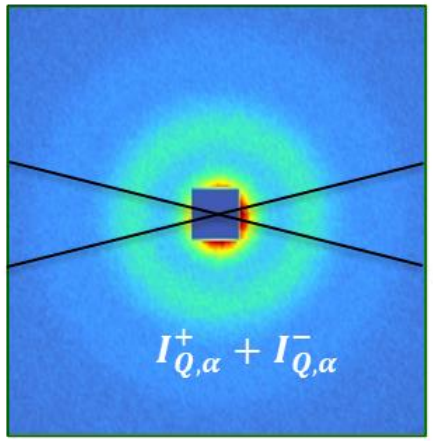
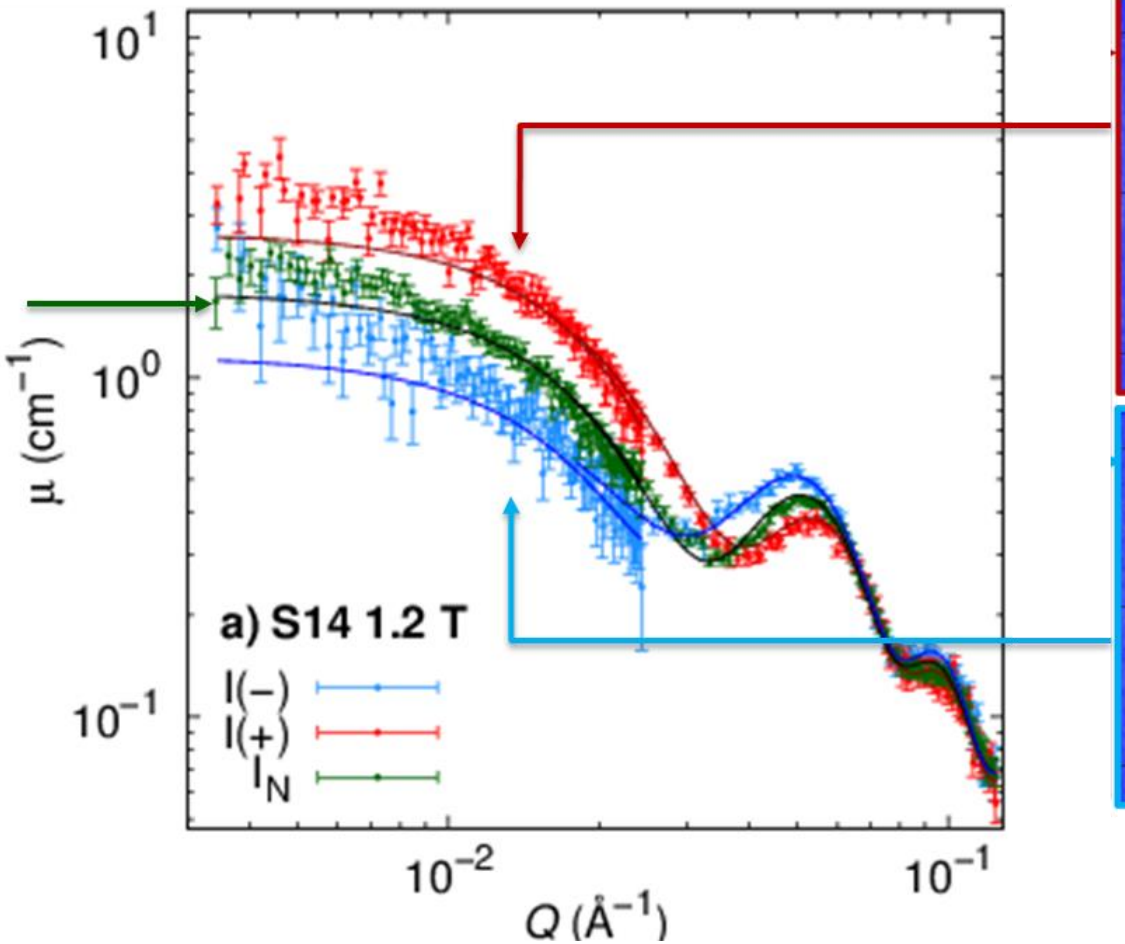
Science and Technology Facilities Council

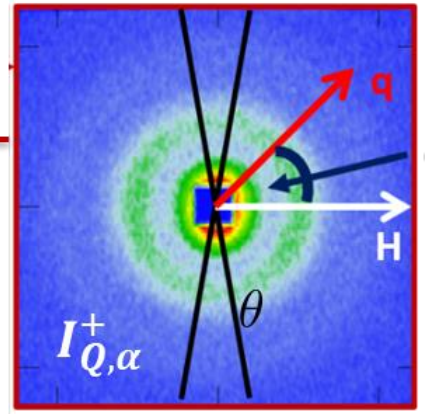
Models and simultaneous constraint fitting in SasView

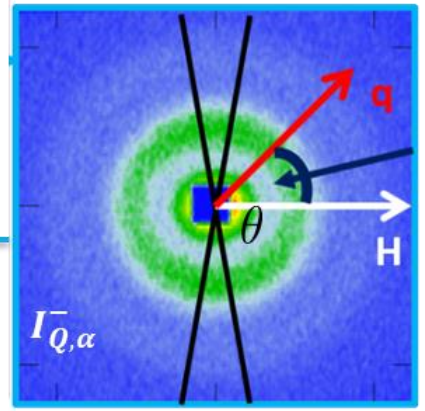
Half-polarised SANS (SANSPOL)











Further analysis:

$$I^{\pm}(\mathbf{q}) \propto |\widetilde{N}|^2 \mp 2\widetilde{N}\widetilde{M}_z \sin^2\theta + |\widetilde{M}_z|^2$$

$$I^{\pm} = N^2 \mp 2NM_z + M_z^2 = (N \mp M_z)^2$$

Example of magnetic nanoparticles

Use sectors from polarised SANS data

Exercise 1: Use unpolarised scattering cross section in the horizontal. This is the purely nuclear, structural scattering.
Determine the particle size, surfactant thickness and the scattering length densities. Use a inorganic core- organic shell model.

Exercise 2: What is the size and scattering length density of the coherently magnetised volume?
Use the vertical sectors Remember that the polarised scattering intensity is
$I^{\pm} = N^2 \mp 2NM_z + M_z^2 = (N \mp M_z)^2$. The additional SLD is magnetic in origin.

Extra:
Try a simultaneous confined fit for both vertical sectors.

To catch further details:
Use an inorganic core- inorganic shell-inorganic shell model.
2) The magnetic volume is 1-2 lattice constants smaller than the actual inorganic particle size.

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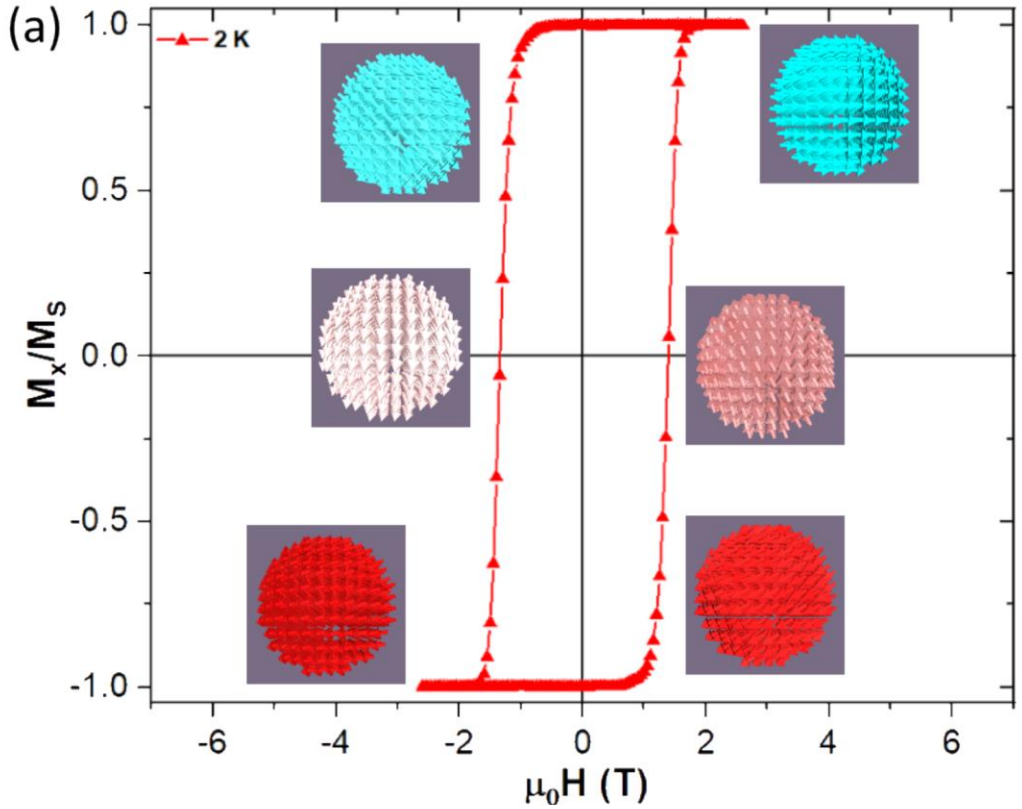
More on reparametrizing models
Tomorrow with Paul

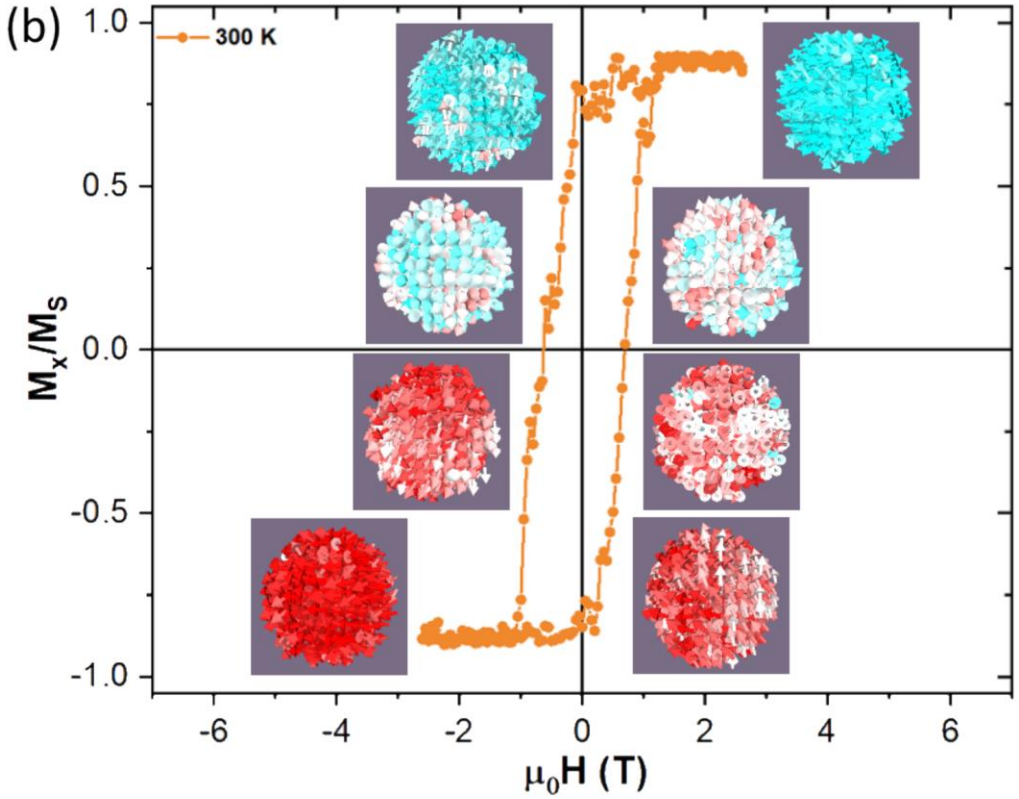


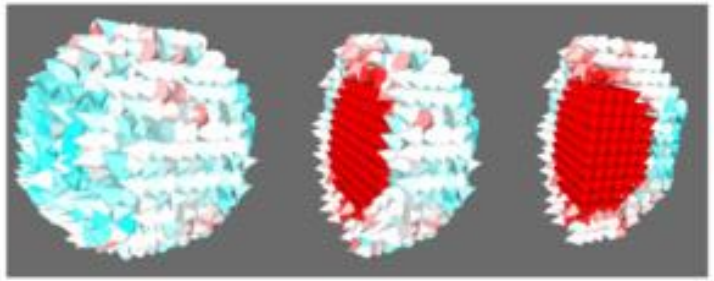
Science and Technology Facilities Council

Generic Scattering Calculator in SasView

Magnetization Dynamics of Nanoparticle with Disordered Shell







Thermal fluctuation and absence of magnetocrystalline anisotropy @ shell

D. Aurélio and J. Vejpravova, Nanomaterials **10**, 1149 (2020)

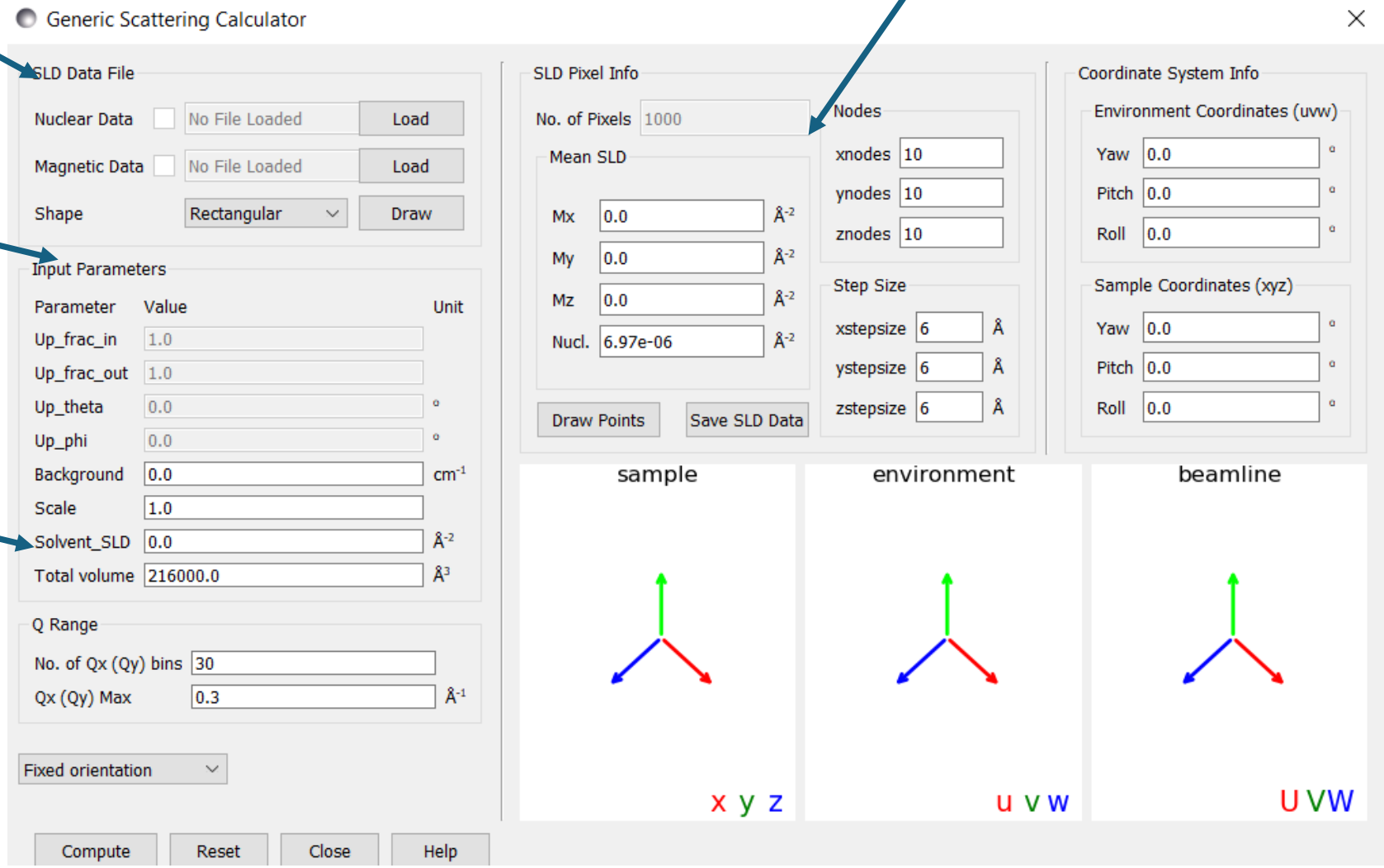
General scattering calculator

Data with SLD profiles (.sld, OMF, PDP, VTK)

Information Panel:
Average SLD
#pixels + step size of rectangular grid

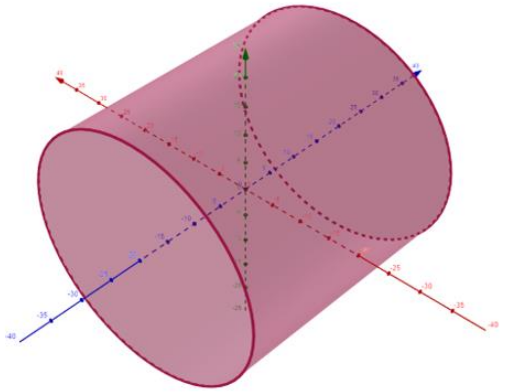
Set Polarisation

Binning + q-range



General scattering calculator: Nuclear-Magnet Scattering

Magnetic Cylinder



Generic Scattering Calculator

SLD Data File

Nuclear Data ☑ mag_cylinder.sld Load

Magnetic Data ☑ mag_cylinder.sld Load

Shape Rectangular Draw

Input Parameters

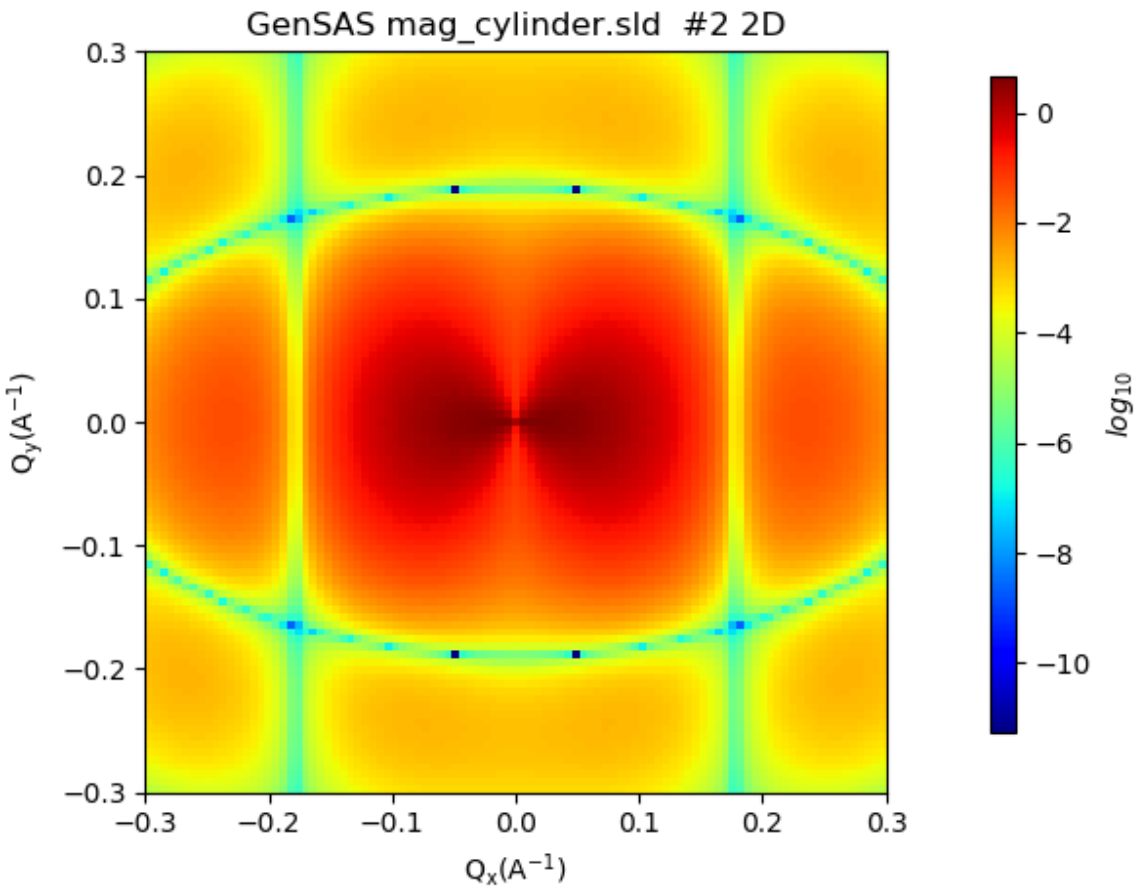
Parameter	Value	Unit
Up_frac_in	0	
Up_frac_out	0	
Up_theta	90	°
Up_phi	0.0	°
Background	0.0	cm^{-1}
Scale	1.0	
Solvent_SLD	0.0	$Å^{-2}$
Total volume	364431.48688046803	$Å^3$

Q Range

No. of Qx (Qy) bins 100

Qx (Qy) Max 0.3 $Å^{-1}$

Compute Reset Close Help



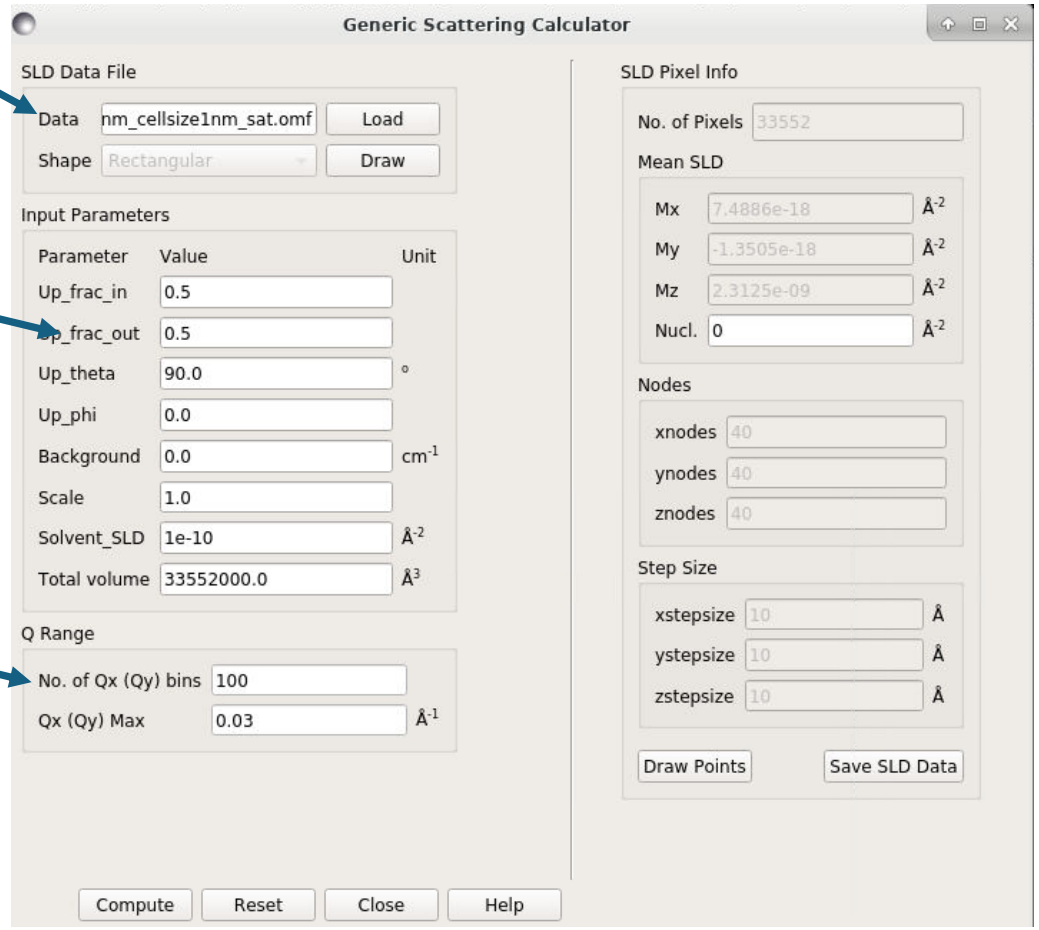
Example in Sasview

General scattering calculator: Polarisation

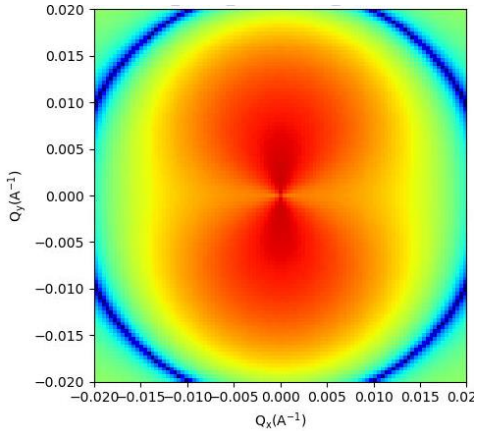
Data with SLD/magnetisation profiles (.sld, .txt, or .omf)

Set Polarisation

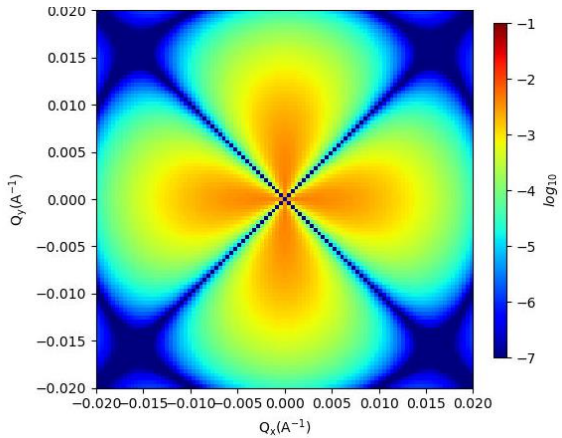
Binning + q-range



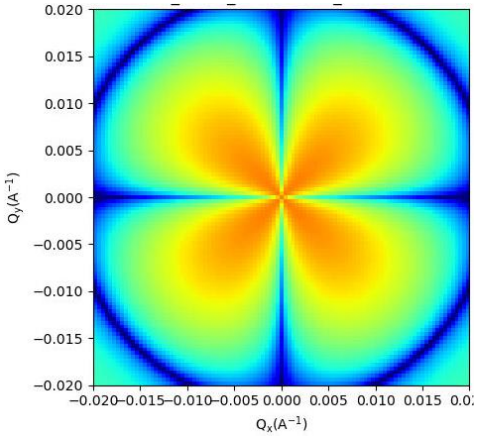
NSF I^{--} $H \perp k_0$ N=0.5 M

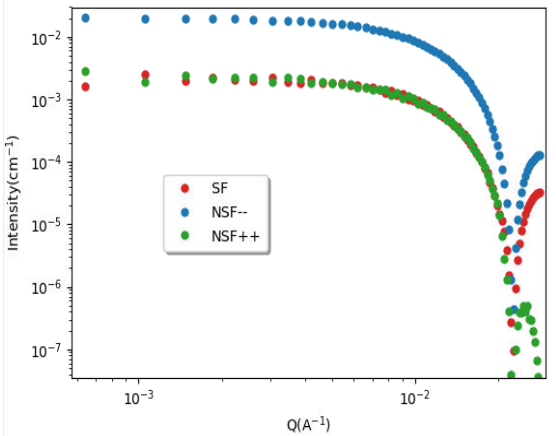


NSF I^{++} $H \perp k_0$ N=0.5 M

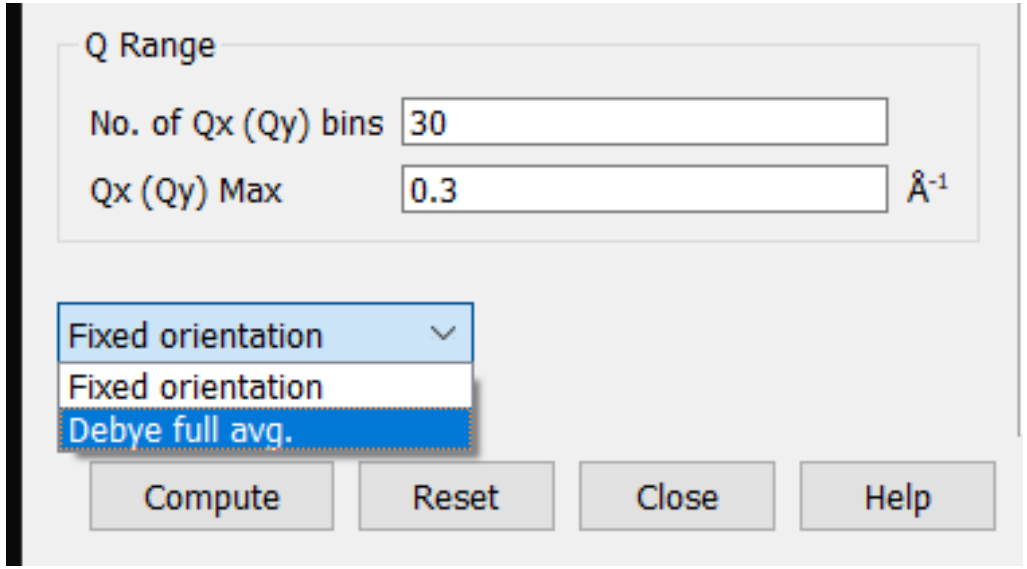


SF I^{+-} $H \perp k_0$





General scattering calculator: Orientation averages



For a generalisation of orientation averages, e.g. for magnetism:
See jupyter notebook

General scattering calculator: Orientation averages

For a generalisation of orientation averages, e.g. for magnetism:
See Generic SAS Calculator Tool documentation with code examples

```
import sys
sys.path.append("filepath to sasview/src")
from sas.sascalc.calculator import sas_gen
```

Loading data from VTK, PDB or SLD files is easily done with:

```
vtkloader = sas_gen.VTKReader()
vtkData = vtkloader.read("filepath to vtk file") # a MagSLD object
```

```
pdbloader = sas_gen.PDBReader()
pdbData = pdbloader.read("filepath to pdb file") # a MagSLD object
```

```
sldloader = sas_gen.SLDReader()
sldData = sldloader.read("filepath to sld file") # a MagSLD object
```

- The x, y and z coordinates of the realspace positions of the pixels
- The nuclear SLD
- The x, y and z components of the magnetic SLD
- The volume of each pixel. Note that grid type data is expected to be on a regular grid - so if *vol_pix* is None the volumes are all taken to be the same.

Model Creation

```
model = sas_gen.GenSAS()
model.set_sld_data(data)
```

```
model.params["Up_theta"] = 90
model.params["Up_phi"] = 45
...
```

Intensity Calculation